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Inventor(s)	Kittelmann; Olaf

Laser source for an ophthalmic surgical system

Abstract

A laser source for an ophthalmic surgical system includes a femtosecond seeder, an amplifier, a femtosecond pulse portion, a nanosecond pulse portion, and one or more switches. The femtosecond seeder generates femtosecond pulses. The amplifier amplifies laser pulses, which include the femtosecond pulses and nanosecond pulses. The amplifier amplifies the laser pulses by amplifying the femtosecond pulses and generating and amplifying the nanosecond pulses. The femtosecond pulse portion alters and outputs the femtosecond pulses, and the nanosecond pulse portion alters and outputs the nanosecond pulses. The switches receive the laser pulses from the amplifier, and direct the laser pulses to the femtosecond pulse portion or the nanosecond pulse portion. In other embodiments, the laser source includes a femtosecond seeder and a nanosecond seeder that generates the nanosecond pulses.

Inventors:	Kittelmann; Olaf (Berlin, DE)
Applicant:	Alcon Inc. (Fribourg, CH)
Family ID:	83355540
Assignee:	ALCON INC. (Fribourg, CH)
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Primary Examiner: Jian; Shirley X

Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to ophthalmic surgical systems, and more particularly to a laser source for an ophthalmic surgical system.

BACKGROUND

(2) Laser-assisted in situ keratomileusis (LASIK) is a type of refractive surgery that reshapes the cornea to correct refractive errors such as myopia, hyperopia, and astigmatism. During surgery, a femtosecond laser photodisrupts corneal tissue to create a flap. The flap is folded back, revealing the stroma. Then, an excimer laser (such as a 193-nanometer laser) ablates the tissue with nanosecond pulses to reshape the corneal stroma to correct the refractive error.

BRIEF SUMMARY

(3) In certain embodiments, a laser source for an ophthalmic surgical system includes a femtosecond seeder, an amplifier, a femtosecond pulse portion, a nanosecond pulse portion, and one or more switches. The femtosecond seeder generates femtosecond pulses. The amplifier amplifies laser pulses, which include the femtosecond pulses and nanosecond pulses. The amplifier amplifies the laser pulses by amplifying the femtosecond pulses and generating and amplifying the nanosecond pulses. The femtosecond pulse portion alters and outputs the femtosecond pulses, and the nanosecond pulse portion alters and outputs the nanosecond pulses. The switches receive the laser pulses from the amplifier, and direct the laser pulses to the femtosecond pulse portion or the nanosecond pulse portion.

(4) Embodiments may include none, one, some, or all of the following features: The laser source further comprises control electronics that: determine if a request is for the femtosecond pulses or the nanosecond pulses; and instruct the switches to direct the laser pulses to the femtosecond pulse portion or the nanosecond pulse portion in accordance with the request. The amplifier further generates the nanosecond pulses by Q-switching. The amplifier includes: an optical switch that operates as a Q-switch; and a pump laser synchronized with the optical switch. The amplifier is a regenerative amplifier, a fiber amplifier, or a multi-pass amplifier chain. The nanosecond pulse portion includes a frequency converter that converts near infrared wavelengths to ultraviolet wavelengths to yield ultraviolet nanosecond pulses. The nanosecond pulse portion includes a nanosecond ablation head that outputs the nanosecond pulses. The femtosecond pulse portion alters and outputs the femtosecond pulses by outputting near infrared femtosecond pulses. The femtosecond pulse portion alters and outputs the femtosecond pulses by outputting ultraviolet femtosecond pulses. The femtosecond pulse portion alters and outputs the femtosecond pulses by outputting near infrared femtosecond pulses and ultraviolet femtosecond pulses. The femtosecond pulse portion may include: a near infrared femtosecond optic head that outputs the near infrared femtosecond pulses; and an ultraviolet femtosecond optic head that outputs the ultraviolet femtosecond pulses. The femtosecond pulse portion includes: a frequency converter that converts near infrared wavelengths to ultraviolet wavelengths to yield ultraviolet femtosecond pulses; and an ultraviolet femtosecond optic head that outputs the ultraviolet femtosecond pulses.

(5) In certain embodiments, a laser source for an ophthalmic surgical system includes femtosecond and nanosecond seeders, an amplifier, a femtosecond pulse portion, a nanosecond pulse portion,

and one or more switches. The seeders generate laser pulses comprising femtosecond pulses and nanosecond pulses. The femtosecond seeder generates the femtosecond pulses, and the nanosecond seeder generates the nanosecond pulses. The amplifier amplifies the femtosecond pulses and the nanosecond pulses. The femtosecond pulse portion alters and outputs the femtosecond pulses, and the nanosecond pulse portion alters and outputs the nanosecond pulses. The switches receive the laser pulses from the amplifier, and direct the laser pulses to the femtosecond pulse portion or the nanosecond pulse portion.

(6) Embodiments may include none, one, some, or all of the following features: The laser source further comprises control electronics that: determine if a request is for the femtosecond pulses or the nanosecond pulses; and instruct the switches to direct the laser pulses to the femtosecond pulse portion or the nanosecond pulse portion in accordance with the request. The amplifier is a regenerative amplifier, a fiber amplifier, or a multi-pass amplifier chain. The nanosecond pulse portion includes a frequency converter that converts near infrared wavelengths to ultraviolet wavelengths to yield ultraviolet nanosecond pulses. The nanosecond pulse portion includes a nanosecond ablation head that outputs the nanosecond pulses. The femtosecond pulse portion alters and outputs the femtosecond pulses by outputting near infrared femtosecond pulses and ultraviolet femtosecond pulses. The femtosecond pulse portion may include: a near infrared femtosecond optic head that outputs the near infrared femtosecond pulses; and an ultraviolet femtosecond optic head that outputs the ultraviolet femtosecond pulses. The femtosecond pulse portion includes: a frequency converter that converts near infrared wavelengths to ultraviolet wavelengths to yield ultraviolet femtosecond pulses; and an ultraviolet femtosecond optic head that outputs the ultraviolet femtosecond pulses.

(7) In certain embodiments, a laser source for an ophthalmic surgical system includes a femtosecond seeder, an amplifier, a femtosecond pulse portion, a nanosecond pulse portion, one or more switches, and control electronics. The femtosecond seeder generates femtosecond pulses. The amplifier amplifies laser pulses, which include the femtosecond pulses and nanosecond pulses. The amplifier amplifies the laser pulses by amplifying the femtosecond pulses and generating and amplifying the nanosecond pulses. The amplifier is a regenerative amplifier, a fiber amplifier, or a multi-pass amplifier chain that generates the nanosecond pulses by Q-switching. The amplifier includes: an optical switch that operates as a Q-switch; and a pump laser synchronized with the optical switch. The femtosecond pulse portion alters and outputs the femtosecond pulses as near infrared femtosecond pulses and ultraviolet femtosecond pulses. The femtosecond pulse portion includes: a near infrared femtosecond optic head that outputs the near infrared femtosecond pulses; a frequency converter that converts near infrared wavelengths to ultraviolet wavelengths to yield ultraviolet femtosecond pulses; and an ultraviolet femtosecond optic head that outputs the ultraviolet femtosecond pulses. The nanosecond pulse portion alters and outputs the nanosecond pulses. The nanosecond pulse portion includes a frequency converter that converts near infrared wavelengths to ultraviolet wavelengths to yield ultraviolet nanosecond pulses, and a nanosecond ablation head that outputs the nanosecond pulses. The switches receive the laser pulses from the amplifier, and direct the laser pulses to the femtosecond pulse portion or the nanosecond pulse portion. The control electronics determine if a request is for the femtosecond pulses or the nanosecond pulses, and instruct the switches to direct the laser pulses to the femtosecond pulse portion or the nanosecond pulse portion in accordance with the request.

(8) In certain embodiments, a laser source for an ophthalmic surgical system includes femtosecond and nanosecond seeders, an amplifier, a femtosecond pulse portion, a nanosecond pulse portion, one or more switches, and control electronics. The seeders generate laser pulses comprising femtosecond pulses and nanosecond pulses. The femtosecond seeder generates the femtosecond pulses, and the nanosecond seeder generates the nanosecond pulses. The amplifier is a regenerative amplifier, a fiber amplifier, or a multi-pass amplifier chain that amplifies the femtosecond pulses and the nanosecond pulses. The femtosecond pulse portion alters and outputs the femtosecond

pulses as near infrared femtosecond pulses and ultraviolet femtosecond pulses. The femtosecond pulse portion includes: a near infrared femtosecond optic head that outputs the near infrared femtosecond pulses; a frequency converter that converts near infrared wavelengths to ultraviolet wavelengths to yield ultraviolet femtosecond pulses; and an ultraviolet femtosecond optic head that outputs the ultraviolet femtosecond pulses. The nanosecond pulse portion alters and outputs the nanosecond pulses. The nanosecond pulse portion includes a frequency converter that converts near infrared wavelengths to ultraviolet wavelengths to yield ultraviolet nanosecond pulses, and a nanosecond ablation head that outputs the nanosecond pulses. The switches receive the laser pulses from the amplifier, and direct the laser pulses to the femtosecond pulse portion or the nanosecond pulse portion. The control electronics determine if a request is for the femtosecond pulses or the nanosecond pulses, and instruct the switches to direct the laser pulses to the femtosecond pulse portion or the nanosecond pulse portion in accordance with the request.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an example of a laser source that provides femtosecond and nanosecond laser pulses for an ophthalmic surgical system, according to certain embodiments;
- (2) FIG. 2 illustrates an example of an amplifier that may be used in the laser source of FIG. 1; and
- (3) FIG. 3 illustrates an example of a method for providing laser pulses for a laser ophthalmic surgical system that may be performed by the laser source of FIG. 1, according to certain embodiments.

DESCRIPTION OF EXAMPLE EMBODIMENTS

- (4) Referring now to the description and drawings, example embodiments of the disclosed apparatuses, systems, and methods are shown in detail. The description and drawings are not intended to be exhaustive or otherwise limit the claims to the specific embodiments shown in the drawings and disclosed in the description. Although the drawings represent possible embodiments, the drawings are not necessarily to scale and certain features may be simplified, exaggerated, removed, or partially sectioned to better illustrate the embodiments.
- (5) Known LASIK systems use femtosecond and excimer lasers to perform the surgery. However, using both types of lasers requires more room and costs more. Moreover, excimer lasers have disadvantages, e.g., they require high voltage, utilize a toxic gas (ArF gas), need complex beam shapers, create ozone, and are bulky. In addition, excimer lasers use radiation with a wavelength that is affected by humidity and oxygen in the air, which affects the ablation rate and requires a humidity-controlled environment and nitrogen purging of the beam paths.
- (6) In certain embodiments, a solid-state laser source provides femtosecond laser pulses for performing flap creation and other intrastromal procedures and nanosecond laser pulses for performing ablation procedures. The laser source uses the same components for generating femtosecond and nanosecond pulses, thus avoiding the problems of using two lasers. Moreover, the laser source is not an excimer laser, thus avoiding the disadvantages of excimer lasers.
- (7) FIG. 1 illustrates an example of a laser source **10** that provides femtosecond and nanosecond laser pulses for an ophthalmic surgical system, according to certain embodiments. In the illustrated example, laser source **10** is a solid-state laser source that includes seeders **20** (which include femtosecond seeder **20a** and/or nanosecond seeder **20b**), amplifier **22**, femtosecond portion **24**, nanosecond portion **26**, switches **28** (e.g., mirrors **28a-e**), and control electronics **50**, coupled (e.g., optically, electrically, and/or mechanically) as shown. Femtosecond portion **24** includes a compressor **30**, a near infrared (NIR) femtosecond portion **24a**, and an ultraviolet (UV) femtosecond portion **24b**, coupled as shown. NIR femtosecond portion **24a** includes a NIR femtosecond optic head **32**. UV femtosecond portion **24b** includes a femtosecond frequency

converter **36** and a UV femtosecond optic head **38**, coupled as shown. Nanosecond portion **26** includes a nanosecond frequency converter **42** and a UV ablation head **44**, coupled as shown. (8) As an example of an overview of operation, seeder(s) **20** (e.g., femtosecond seeder **20a** and optionally nanosecond seeder **20b**) generate laser pulses, and amplifier **22** amplifies the laser pulses. Switches **28** receive the laser pulses from amplifier **22** and direct the laser pulses to femtosecond pulse portion **24** or nanosecond pulse portion **26**. In certain embodiments, control electronics **50** may determine if a request is for femtosecond or nanosecond pulses, and instruct switches **28** to direct the laser pulses to the femtosecond **24** or nanosecond **26** pulse portion in accordance with the request. Femtosecond pulse portion **24** outputs femtosecond pulses, or nanosecond pulse portion **26** outputs nanosecond pulses.

(9) Turning to the details of the example, laser source **10** includes seeders **20**, e.g., femtosecond seeder **20a** and/or nanosecond seeder **20b**. In certain embodiments, laser source **10** includes femtosecond and nanosecond seeders **20** that generate femtosecond and nanosecond pulses. In other embodiments, laser source **10** includes a femtosecond seeder **20a**, but not a nanosecond seeder **20b**, so only femtosecond pulses are generated in a seeder. Amplifier **22** (which may be a regenerative amplifier) can generate the nanosecond pulses by, e.g., Q-switching.

(10) Amplifier **22** may comprise one or more cascaded amplifiers, and may comprise a regenerative amplifier, a fiber amplifier, or a multi-pass amplifier chain. If laser source **10** includes femtosecond and nanosecond seeders **20**, a fiber amplifier may be used. If laser source **10** does not include a nanosecond seeder **20b**, a regenerative amplifier can generate the nanosecond pulses by Q-switching. The regenerative amplifier may have the gain medium (e.g., a solid-state medium) in an optical resonator where pulses make multiple round-trip passes. An optical switch controls the number of passes, allowing for a large number of passes that yield high amplification. An example of amplifier **22** is described in more detail with reference to FIG. 2.

(11) In the example, the pulses from seeder(s) **20** have near infrared (NIR) wavelengths, e.g., 1000 to 1700 nanometers (nm). Wavelength/frequency converters transform the NIR wavelengths to UV wavelengths, e.g., 340 to 360 nm for femtosecond pulses and 200 to 220 nm, such as 204 to 216 nm (deep UV or DUV), for nanosecond pulses. Frequency converters may include cascaded stages of optical non-linear crystals and may be designed to match fundamental and required target wavelengths by, e.g., generating the Third Harmonic of the fundamental femtosecond pulses or generating Fifth Harmonic of the fundamental nanosecond pulses.

(12) Nanosecond portion **26** includes nanosecond frequency converter **42** and UV nanosecond ablation head **44**. Frequency converter **42** transforms the nanosecond pulses from NIR wavelengths (e.g., 1020 to 1080 nm) to deep UV wavelengths (e.g., 204 to 216 nm). UV nanosecond ablation head **44** outputs nanosecond pulses, which may be used to ablate tissue such as corneal tissue.

(13) Femtosecond portion **24** provides NIR and UV femtosecond pulses, which may be used to treat tissue, such as corneal tissue, by photodisruption. In the example, compressor **30** compresses laser pulses from amplifier **22** to femtosecond pulse durations. NIR femtosecond portion **24a** includes a NIR femtosecond optic head **32** that outputs NIR femtosecond pulses. UV femtosecond portion **24b** includes femtosecond frequency converter **36** and UV femtosecond optic head **38**. Frequency converter **36** transforms the femtosecond pulses from NIR wavelengths (e.g., 1020 to 1080 nm) to UV wavelengths (e.g., 340 to 360 nm) via, e.g., frequency tripling. UV femtosecond optic head **38** outputs UV femtosecond pulses.

(14) Switches **28** such as mirrors **28a-e** direct laser pulses to components in response to instructions from control electronics **50**. A switch **28** may be any suitable optical elements that can switch laser pulses, e.g., a mirror, Pockels cell and Polarizer, acousto-optic modulator, micro-electro-mechanical system, or galvo mirror.

(15) Control electronics **50** sends instructions to control the components of laser source **10**. In certain embodiments, control electronics **50** may determine if a request is for femtosecond or nanosecond pulses, and instruct switches **28** to direct the laser pulses to the femtosecond **24** or

nanosecond 26 pulse portion in accordance with the request.

(16) FIG. 2 illustrates an example of an amplifier 22 that may be used in laser source 10 of FIG. 1. In the example, amplifier 22 is a regenerative amplifier that includes a pump laser 110, an optical resonator 112, and an input/output (I/O) 116, coupled (e.g., optically) as shown. Seed laser 110 includes a laser diode 120 and pump optics 122, coupled as shown. Optical resonator 112 includes an optical switch 114, mirrors 126, 130, 132, 134 and an amplifier crystal 128, coupled as shown. Optical switch 114 includes a mirror 134, an electro-optic device such as a Pockels cell 136, a quarter-wave plate 140, and a thin-film polarizer (TFP) 142, coupled as shown. In another embodiment, an acousto-optic modulator may be used as the optical switch. Input/output 116 includes an input 144, an output 146, a TFP 150, a half-wave plate 140, and a Faraday rotator 154, coupled as shown.

(17) As an overview of operation, input 144 provides seed pulses to optical resonator 112 via input/output (I/O) 116. Optical resonator 112 amplifies laser pulses when the pulses make round trips through amplifier crystal 128 within the resonator. Optical switch 114 controls the entrance and exit of a pulse into and out of optical resonator 112, which controls the amplification of the pulse. Input/output 116 directs pulses to exit amplifier 22 via output 146.

(18) Turning to details of the example, pump laser 110 provides energy to activate (pump) amplifier crystal 128 to increase the energy of the circulating pulses by several orders of magnitude. A greater number of round trips results in more amplification. Mirrors 126, 130, 132, 134 of optical resonator 112 direct pulses through amplifier crystal 128 and to optical switch 114. A mirror may be any suitable optical element that reflects or otherwise directs laser pulses. Amplifier crystal 128 amplifies laser pulses and may be any suitable laser crystal, e.g., Ytterbium or Neodymium doped materials.

(19) Optical switch 114 may be an electro-optic or acousto-optic switch. Pockels cell 136, quarter-wave plate 140, and thin-film polarizer (TFP) 142 operate as an optical switch to transmit or reflect pulses, in order to switch pulses between optical resonator 112 and input/output 116. This controls the amount of amplification. Input 144 of input/output 116 feeds pulses into amplifier 22, and output 146 allows pulses out of amplifier 22. Faraday rotator 154 and half-wave plate 152 operate as an optical diode to separate the input and output pulses.

(20) In certain embodiments, such as where there is no nanosecond seeder, amplifier 22 (which may be a regenerative amplifier) can generate the nanosecond pulses by, e.g., Q-switching. Amplifier 22 operates as a Q-switched laser resonator, where optical switch 114 acts as Q-switch. Q-switching modulates the intracavity losses and the Q factor of the laser resonator. Q-switching can be supported by pump laser 110 synchronized with optical switch 114. Pump laser 110 may operate as a pulsed pump source or a continuous wave pump laser.

(21) FIG. 3 illustrates an example of a method for providing laser pulses for a laser ophthalmic surgical system that may be performed by the laser source of FIG. 1, according to certain embodiments. The method starts at step 210, where one or more seeders generate laser pulses. In certain embodiments, the laser source includes femtosecond and nanosecond seeders that generate femtosecond and nanosecond pulses. In other embodiments, the laser source includes a femtosecond seeder, but not a nanosecond seeder, so only femtosecond pulses are generated, and an amplifier generates the nanosecond pulses.

(22) An amplifier amplifies the laser pulses at step 212. If the laser source includes only a femtosecond seeder, but not a nanosecond seeder, the amplifier also generates the nanosecond pulses via Q-switching. The laser source may be requested to provide femtosecond or nanosecond pulses at step 214. In certain embodiments, control electronics determine if the request is for femtosecond or nanosecond pulses, and instruct switches to direct the laser pulses to a femtosecond or nanosecond pulse portion in accordance with the request.

(23) If nanosecond pulses are requested at step 214, the method proceeds to step 220, where the laser pulses are directed to the nanosecond pulse portion. A frequency converter converts near

infrared (NIR) wavelengths to ultraviolet (UV) wavelengths to yield UV nanosecond pulses at step 222. A nanosecond ablation head outputs the nanosecond pulses at step 224. The pulses may be used to ablate tissue. The method then ends.

(24) If femtosecond pulses are requested at step 214, the method proceeds to step 230, where the laser pulses are directed to the femtosecond pulse portion. A compressor compresses the NIR femtosecond pulses at step 234 to a femtosecond pulse duration. The laser source may be requested to provide UV or NIR femtosecond pulses at step 232. In certain embodiments, control electronics determine if the request is for UV or NIR femtosecond pulses, and instruct switches to direct the laser pulses to the UV or NIR femtosecond pulse portion in accordance with the request.

(25) If UV femtosecond pulses are requested at step 234, the method proceeds to step 240, where the laser pulses are directed to the UV femtosecond pulse portion. A frequency converter converts near infrared wavelengths to ultraviolet wavelengths to yield UV femtosecond pulses at step 242. A UV femtosecond optic head outputs the UV femtosecond pulses at step 244. The pulses may be used to photodisrupt tissue. The method then ends.

(26) If NIR femtosecond pulses are requested at step 234, the method proceeds to step 250, where the laser pulses are directed to the NIR femtosecond pulse portion. A NIR femtosecond optic head outputs the NIR femtosecond pulses at step 254. The pulses may be used to photodisrupt tissue. The method then ends.

(27) A component (such as control electronics 50) of the systems and apparatuses disclosed herein may include an interface, logic, and/or memory, any of which may include computer hardware and/or software. An interface can receive input to the component and/or send output from the component, and is typically used to exchange information between, e.g., software, hardware, peripheral devices, users, and combinations of these. A user interface is a type of interface that a user can utilize to communicate with (e.g., send input to and/or receive output from) a computer. Examples of user interfaces include a display, Graphical User Interface (GUI), touchscreen, keyboard, mouse, gesture sensor, microphone, and speakers.

(28) Logic can perform operations of the component. Logic may include one or more electronic devices that process data, e.g., execute instructions to generate output from input. Examples of such an electronic device include a computer, processor, microprocessor (e.g., a Central Processing Unit (CPU)), and computer chip. Logic may include computer software that encodes instructions capable of being executed by an electronic device to perform operations. Examples of computer software include a computer program, application, and operating system.

(29) A memory can store information and may comprise tangible, computer-readable, and/or computer-executable storage medium. Examples of memory include computer memory (e.g., Random Access Memory (RAM) or Read Only Memory (ROM)), mass storage media (e.g., a hard disk), removable storage media (e.g., a Compact Disk (CD) or Digital Video or Versatile Disk (DVD)), database, network storage (e.g., a server), and/or other computer-readable media. Particular embodiments may be directed to memory encoded with computer software.

(30) Although this disclosure has been described in terms of certain embodiments, modifications (such as changes, substitutions, additions, omissions, and/or other modifications) of the embodiments will be apparent to those skilled in the art. Accordingly, modifications may be made to the embodiments without departing from the scope of the invention. For example, modifications may be made to the systems and apparatuses disclosed herein. The components of the systems and apparatuses may be integrated or separated, or the operations of the systems and apparatuses may be performed by more, fewer, or other components, as apparent to those skilled in the art. As another example, modifications may be made to the methods disclosed herein. The methods may include more, fewer, or other steps, and the steps may be performed in any suitable order, as apparent to those skilled in the art.

(31) To aid the Patent Office and readers in interpreting the claims, Applicants note that they do not intend any of the claims or claim elements to invoke 35 U.S.C. § 112(f), unless the words “means

for” or “step for” are explicitly used in the particular claim. Use of any other term (e.g., “mechanism,” “module,” “device,” “unit,” “component,” “element,” “member,” “apparatus,” “machine,” “system,” “processor,” or “controller”) within a claim is understood by the applicants to refer to structures known to those skilled in the relevant art and is not intended to invoke 35 U.S.C. § 112(f).

Claims

1. A laser source for an ophthalmic surgical system, the laser source comprising: a femtosecond seeder configured to generate a plurality of femtosecond pulses; and an amplifier configured to: generate a plurality of nanosecond pulses; and amplify a plurality of laser pulses, the plurality of laser pulses comprising the plurality of femtosecond pulses and the plurality of nanosecond pulses, the amplifier configured to amplify the plurality of laser pulses by: amplifying the plurality of femtosecond pulses generated by the femtosecond seeder; and amplifying the plurality of nanosecond pulses generated by the amplifier; a femtosecond pulse portion configured to alter and output the plurality of femtosecond pulses; a nanosecond pulse portion configured to alter and output the plurality of nanosecond pulses; and one or more switches configured to: receive the plurality of laser pulses from the amplifier; and direct the plurality of laser pulses to the femtosecond pulse portion or the nanosecond pulse portion.
2. The laser source of claim 1, further comprising control electronics configured to: determine if a request is for the plurality of femtosecond pulses or the plurality of nanosecond pulses; and instruct the one or more switches to direct the plurality of laser pulses to the femtosecond pulse portion or the nanosecond pulse portion in accordance with the request.
3. The laser source of claim 1, the amplifier further configured to generate the plurality of nanosecond pulses by Q-switching.
4. The laser source of claim 3, the amplifier comprising: an optical switch configured to operate as a Q-switch; and a pump laser synchronized with the optical switch.
5. The laser source of claim 1, the amplifier selected from the following: a regenerative amplifier, a fiber amplifier, or a multi-pass amplifier chain.
6. The laser source of claim 1, the nanosecond pulse portion comprising: a frequency converter configured to convert near infrared wavelengths to ultraviolet wavelengths to yield a plurality of ultraviolet nanosecond pulses.
7. The laser source of claim 1, the nanosecond pulse portion comprising: a nanosecond ablation head configured to output the plurality of nanosecond pulses.
8. The laser source of claim 1, the femtosecond pulse portion configured to alter and output the plurality of femtosecond pulses by outputting a plurality of near infrared femtosecond pulses.
9. The laser source of claim 1, the femtosecond pulse portion configured to alter and output the plurality of femtosecond pulses by outputting a plurality of ultraviolet femtosecond pulses.
10. The laser source of claim 1, the femtosecond pulse portion configured to alter and output the plurality of femtosecond pulses by: outputting a plurality of near infrared femtosecond pulses; and outputting a plurality of ultraviolet femtosecond pulses.
11. The laser source of claim 10, the femtosecond pulse portion comprising: a near infrared femtosecond optic head configured to output the plurality of near infrared femtosecond pulses; and an ultraviolet femtosecond optic head configured to output the plurality of ultraviolet femtosecond pulses.
12. The laser source of claim 1, the femtosecond pulse portion comprising: a frequency converter configured to convert near infrared wavelengths to ultraviolet wavelengths to yield a plurality of ultraviolet femtosecond pulses; and an ultraviolet femtosecond optic head configured to output the plurality of ultraviolet femtosecond pulses.
13. A laser source for an ophthalmic surgical system, the laser source comprising: a femtosecond

seeder configured to generate a plurality of femtosecond pulses; an amplifier comprising: an optical switch configured to operate as a Q-switch; and a pump laser synchronized with the optical switch; the amplifier configured to: generate a plurality of nanosecond pulses by q-switching; and amplify a plurality of laser pulses, the plurality of laser pulses comprising the plurality of femtosecond pulses and the plurality of nanosecond pulses, the amplifier configured to amplify the plurality of laser pulses by: amplifying the plurality of femtosecond pulses generated by the femtosecond seeder; and amplifying the plurality of nanosecond pulses generated by the amplifier; a femtosecond pulse portion configured to alter and output the plurality of femtosecond pulses; a nanosecond pulse portion configured to alter and output the plurality of nanosecond pulses; and one or more switches configured to: receive the plurality of laser pulses from the amplifier; and direct the plurality of laser pulses to the femtosecond pulse portion or the nanosecond pulse portion.

14. The laser source of claim 13, further comprising control electronics configured to: determine if a request is for the plurality of femtosecond pulses or the plurality of nanosecond pulses; and instruct the one or more switches to direct the plurality of laser pulses to the femtosecond pulse portion or the nanosecond pulse portion in accordance with the request.

15. The laser source of claim 13, the amplifier selected from the following: a regenerative amplifier, a fiber amplifier, or a multi-pass amplifier chain.

16. The laser source of claim 13, the nanosecond pulse portion comprising: a frequency converter configured to convert near infrared wavelengths to ultraviolet wavelengths to yield a plurality of ultraviolet nanosecond pulses.

17. The laser source of claim 13, the nanosecond pulse portion comprising: a nanosecond ablation head configured to output the plurality of nanosecond pulses.

18. The laser source of claim 13, the femtosecond pulse portion configured to output the plurality of femtosecond pulses by: outputting a plurality of near infrared femtosecond pulses; and outputting a plurality of ultraviolet femtosecond pulses.

19. The laser source of claim 18, the femtosecond pulse portion comprising: a near infrared femtosecond optic head configured to output the plurality of near infrared femtosecond pulses; and an ultraviolet femtosecond optic head configured to output the plurality of ultraviolet femtosecond pulses.

20. The laser source of claim 13, the femtosecond pulse portion comprising: a frequency converter configured to convert near infrared wavelengths to ultraviolet wavelengths to yield a plurality of ultraviolet femtosecond pulses; and an ultraviolet femtosecond optic head configured to output the plurality of ultraviolet femtosecond pulses.

21. A laser source for an ophthalmic surgical system, the laser source comprising: a femtosecond seeder configured to generate a plurality of femtosecond pulses; and an amplifier configured to amplify a plurality of laser pulses, the plurality of laser pulses comprising the plurality of femtosecond pulses and a plurality of nanosecond pulses, the amplifier comprising a regenerative amplifier comprising: an optical switch configured to operate as a Q-switch; and a pump laser synchronized with the optical switch, the amplifier configured to amplify the plurality of laser pulses by: amplifying the plurality of femtosecond pulses generated by the femtosecond seeder; generating the plurality of nanosecond pulses by Q-switching; and amplifying the plurality of nanosecond pulses generated by the amplifier; a femtosecond pulse portion configured to alter and output the plurality of femtosecond pulses by outputting a plurality of near infrared femtosecond pulses and a plurality of ultraviolet femtosecond pulses, the femtosecond pulse portion comprising: a near infrared femtosecond optic head configured to output the plurality of near infrared femtosecond pulses; a frequency converter configured to convert near infrared wavelengths to ultraviolet wavelengths to yield a plurality of ultraviolet femtosecond pulses; and an ultraviolet femtosecond optic head configured to output the plurality of ultraviolet femtosecond pulses; a nanosecond pulse portion configured to alter and output the plurality of nanosecond pulses, the

nanosecond pulse portion comprising: a frequency converter configured to convert near infrared wavelengths to ultraviolet wavelengths to yield a plurality of ultraviolet nanosecond pulses; and a nanosecond ablation head configured to output the plurality of nanosecond pulses; one or more switches configured to: receive the plurality of laser pulses from the amplifier; and direct the plurality of laser pulses to the femtosecond pulse portion or the nanosecond pulse portion; and control electronics configured to: determine if a request is for the plurality of femtosecond pulses or the plurality of nanosecond pulses; and instruct the one or more switches to direct the plurality of laser pulses to the femtosecond pulse portion or the nanosecond pulse portion in accordance with the request.

22. A laser source for an ophthalmic surgical system, the laser source comprising: a femtosecond seeder configured to generate a plurality of femtosecond pulses; an amplifier comprising: an optical switch configured to operate as a Q-switch; and a pump laser synchronized with the optical switch; the amplifier configured to: generate a plurality of nanosecond pulses by q-switching; and amplify a plurality of laser pulses, the plurality of laser pulses comprising the plurality of femtosecond pulses and the plurality of nanosecond pulses, the amplifier configured to amplify the plurality of laser pulses by: amplifying the plurality of femtosecond pulses generated by the femtosecond seeder; and amplifying the plurality of nanosecond pulses generated by the amplifier, the amplifier comprising a regenerative amplifier, a fiber amplifier, or a multi-pass amplifier chain; a femtosecond pulse portion configured to alter and output the plurality of femtosecond pulses by outputting a plurality of near infrared femtosecond pulses and a plurality of ultraviolet femtosecond pulses, the femtosecond pulse portion comprising: a near infrared femtosecond optic head configured to output the plurality of near infrared femtosecond pulses; a frequency converter configured to convert near infrared wavelengths to ultraviolet wavelengths to yield a plurality of ultraviolet femtosecond pulses; and an ultraviolet femtosecond optic head configured to output the plurality of ultraviolet femtosecond pulses; a nanosecond pulse portion configured to alter and output the plurality of nanosecond pulses, the nanosecond pulse portion comprising: a frequency converter configured to convert near infrared wavelengths to ultraviolet wavelengths to yield a plurality of ultraviolet nanosecond pulses; and a nanosecond ablation head configured to output the plurality of nanosecond pulses; one or more switches configured to: receive the plurality of laser pulses from the amplifier; and direct the plurality of laser pulses to the femtosecond pulse portion or the nanosecond pulse portion; and control electronics configured to: determine if a request is for the plurality of femtosecond pulses or the plurality of nanosecond pulses; and instruct the one or more switches to direct the plurality of laser pulses to the femtosecond pulse portion or the nanosecond pulse portion in accordance with the request.
